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Training Data for Large Language Models

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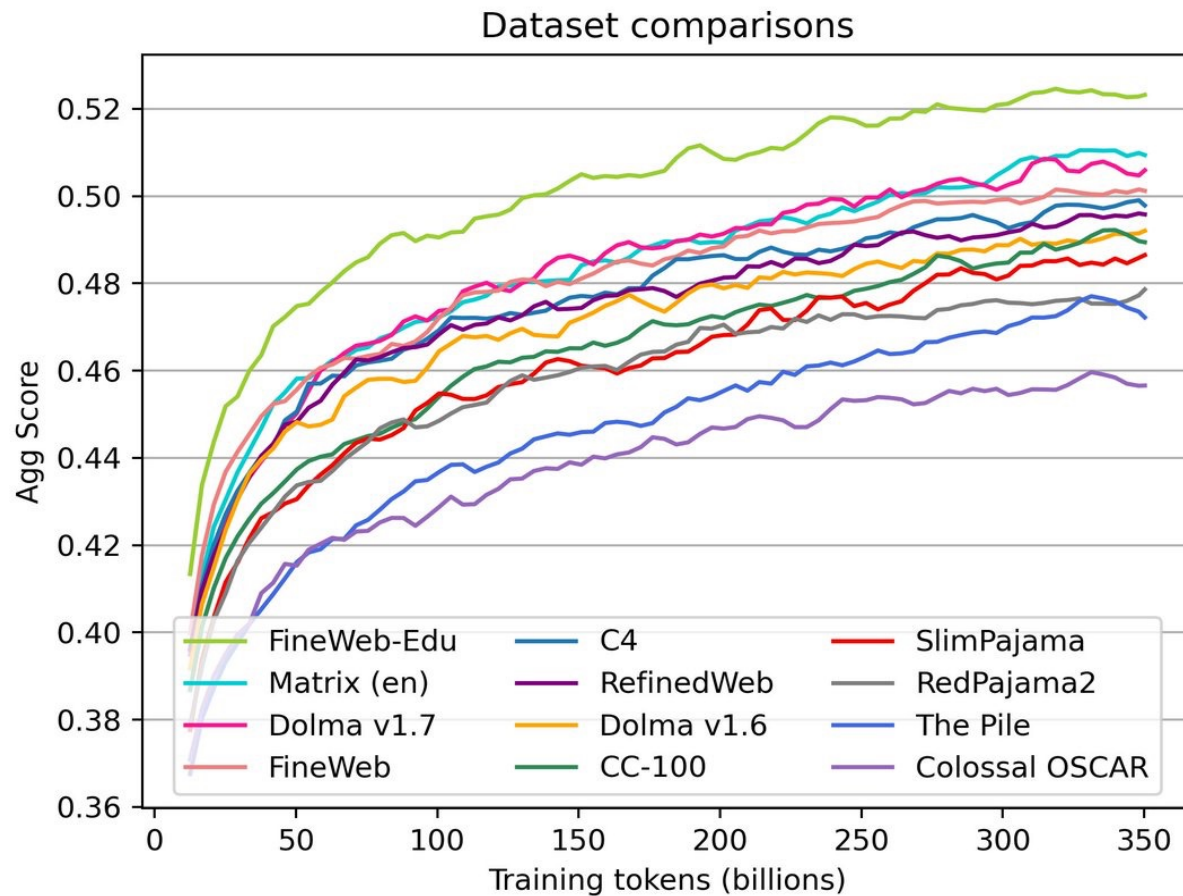
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Why Training Data Matters

- The fossil fuels for training LLMs
 - data drives what they *learn* and how they *behave*!
- To develop broad capabilities (e.g., language understanding, reasoning, world knowledge), models must be trained on diverse and extensive text corpora.
 - e.g., science, fiction, encyclopedias, emails, etc.
 - Help ensure generalisation and robustness
- Poorly curated data can introduce bias, misinformation, or safety risks.

Why Training Data Matters



- As the amount of training data increases, high-quality datasets yield significantly better performance than low-quality ones.
- Investing effort in building or selecting a higher-quality corpus yields better model performance than simply increasing token count.

Key Datasets for Training LLMs

- **Common Crawl**
 - The raw foundation for many datasets, including C4 and parts of GPT-3.
- **Colossal Clean Crawled Corpus (C4)**
 - A cleaned and filtered version of Common Crawl used by T5 and others.
- **WebText & OpenWebText**
 - Curated web content based on Reddit links; more targeted than Common Crawl.
- **The Pile**
 - A modular, high-quality open-source dataset that aggregates diverse sources including books, GitHub, academic papers, etc.
- **GPT-3 Dataset**
 - A proprietary mix that draws from many of the above (Common Crawl, WebText, books, Wikipedia, code, etc.).

Common Crawl



The Data ▾

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Common Crawl
maintains a free, open
repository of web
crawl data that can be
used by anyone.

Common Crawl is a 501(c)(3) non-profit founded in

<https://commoncrawl.org/>

- Nonprofit project providing open-access web crawl data (~320 TB)
- Maintains regular snapshots of the internet, freely available to anyone
- A standard data source for LLMs (e.g., T5, GPT-3, Gopher)
- Popular due to scale, openness, and ease of use

Common Crawl

- Used in major models: GPT-3, T5, Gopher, etc.
- Contains a wide range of contents: *blogs, forums, news*, etc.
- **Strengths:** massive scale, easily accessible
- **Limitations:** noisy, contains spam, boilerplate, duplicated content
- **Not curated:** Requires significant filtering for quality, ethics, and relevance

WebText and OpenWebText

- **Background**
 - Early LLMs used datasets based on news, Wikipedia, or fiction
 - Common Crawl was large but noisy (e.g., spam, boilerplate)
- **WebText (OpenAI)**
 - URLs from Reddit posts with ≥ 3 upvotes
 - Curated for high-quality, diverse content (*excluding Wikipedia* to allow fair benchmark evaluation)
 - Used to train GPT-2 (~40 GB);
- **OpenWebText (Community):**
 - Community-built replica of WebText (since WebText was never released)
 - URLs from Reddit submissions, filtered with Facebook's fastText (English only)
 - Deduplicated content (~38 GB)
- **Insight:** Curation improves quality but inherits Reddit user bias (e.g., young, male, Western)

Colossal Clean Crawled Corpus (C4)



Colossal Clean Crawled Corpus (C4)

- **Filtered** version of April 2019 Common Crawl (keeps ~10%)
- **Size:** ~156B tokens (~806 GB of English-only text)
- Used to train models like T5
- **Filtering criteria**
 - Removed: profanity, code, boilerplate, very short documents
 - Language filtering using langdetect

Filter decisions affect which voices and topics remain.

C4 Compared to Raw Common Crawl

Feature	Common Crawl	C4
Size	~320 TB	~806 GB
Curation	Minimal	Heavily filtered
Language	Multilingual	English-only
Quality	Noisy, redundant	Cleaner, more structured

C4 is a cleaner, smaller, English-only subset of Common Crawl, designed to improve data quality for model training at the cost of reduced diversity.

GPT-3 Dataset

- **Source:** Built from 2016–2019 Common Crawl snapshots
- Trained a binary classifier to distinguish *WebText-like* vs. *Common Crawl* content
 - Helps retain higher-quality, curated web content
- Content Selection & Cleanup:
 - Applied fuzzy matching (13-gram overlap)
 - Removed content if it appeared in fewer than 10 documents
 - Excluded test data from common benchmark datasets (to avoid data leakage)
- Expanded the diversity of the data sources
 - Included: WebText2, Books1, Books2, Wikipedia

GPT-3 Dataset

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

Table 2.2: Datasets used to train GPT-3. “Weight in training mix” refers to the fraction of examples during training that are drawn from a given dataset, which we intentionally do not make proportional to the size of the dataset. As a result, when we train for 300 billion tokens, some datasets are seen up to 3.4 times during training while other datasets are seen less than once.

The Pile

- Developed by EleutherAI, an open-source AI research group.
- **Goal:** (1) Build a high-quality, diverse dataset for training open LLMs; (2) Improve representation of underrepresented domains
- **Content:** 825 GB of English text from 22 sources including:
 - Academic papers, GitHub, StackExchange, books, medical/legal corpora.
 - Balances quality and diversity by combining technical, scientific, and informal sources, e.g. biomedical, programming, law, online forums
- **Used in:** GPT-Neo, GPT-J, Pythia, and other open models.

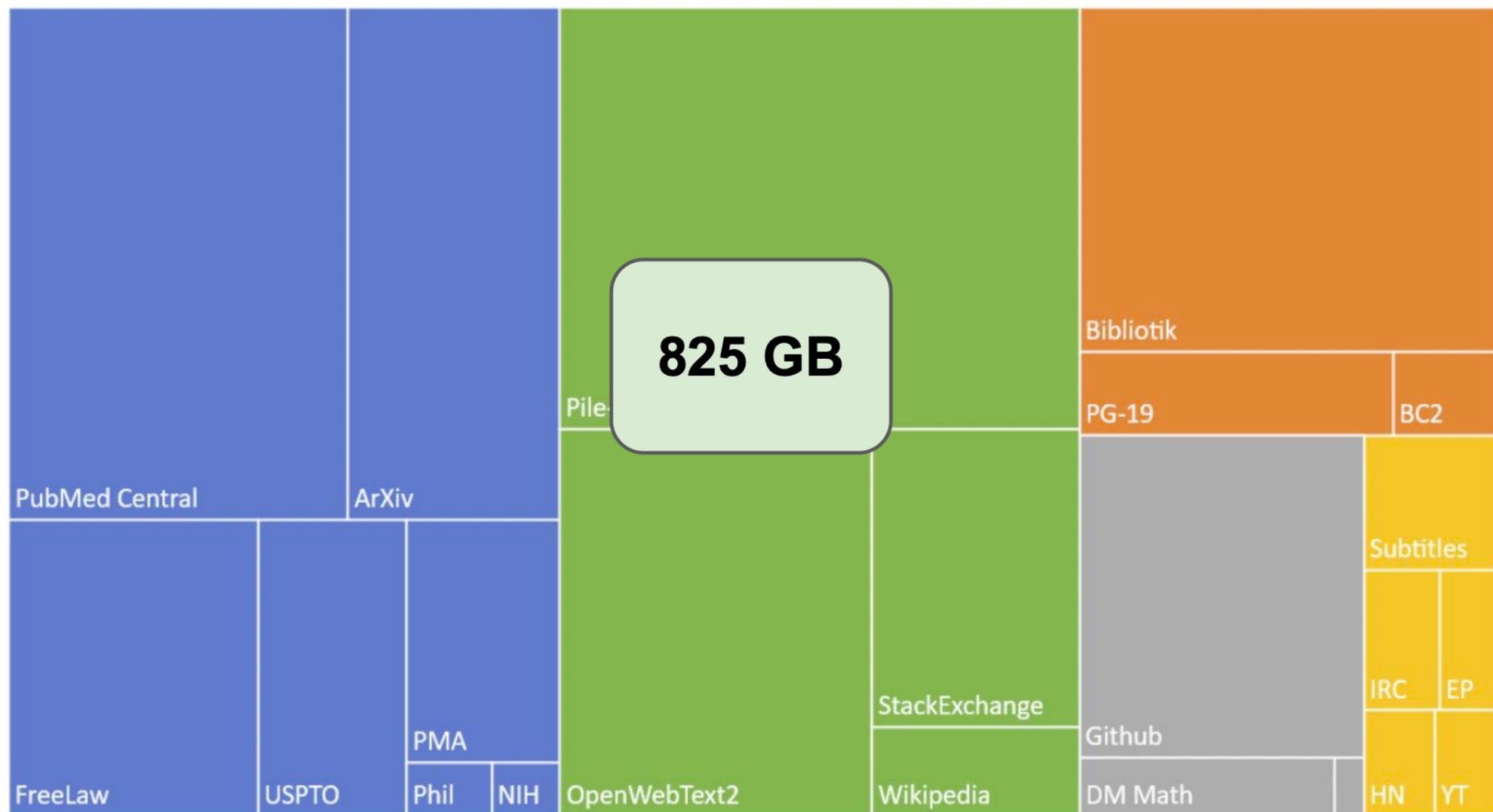
The Pile

Component	Raw Size	Weight	Epochs	Effective Size	Mean Document Size
Pile-CC	227.12 GiB	18.11%	1.0	227.12 GiB	4.33 KiB
PubMed Central	90.27 GiB	14.40%	2.0	180.55 GiB	30.55 KiB
Books3 [†]	100.96 GiB	12.07%	1.5	151.44 GiB	538.36 KiB
OpenWebText2	62.77 GiB	10.01%	2.0	125.54 GiB	3.85 KiB
ArXiv	56.21 GiB	8.96%	2.0	112.42 GiB	46.61 KiB
Github	95.16 GiB	7.59%	1.0	95.16 GiB	5.25 KiB
FreeLaw	51.15 GiB	6.12%	1.5	76.73 GiB	15.06 KiB
Stack Exchange	32.20 GiB	5.13%	2.0	64.39 GiB	2.16 KiB
USPTO Backgrounds	22.90 GiB	3.65%	2.0	45.81 GiB	4.08 KiB
PubMed Abstracts	19.26 GiB	3.07%	2.0	38.53 GiB	1.30 KiB
Gutenberg (PG-19) [†]	10.88 GiB	2.17%	2.5	27.19 GiB	398.73 KiB
OpenSubtitles [†]	12.98 GiB	1.55%	1.5	19.47 GiB	30.48 KiB
Wikipedia (en) [†]	6.38 GiB	1.53%	3.0	19.13 GiB	1.11 KiB
DM Mathematics [†]	7.75 GiB	1.24%	2.0	15.49 GiB	8.00 KiB
Ubuntu IRC	5.52 GiB	0.88%	2.0	11.03 GiB	545.48 KiB
BookCorpus2	6.30 GiB	0.75%	1.5	9.45 GiB	369.87 KiB
EuroParl [†]	4.59 GiB	0.73%	2.0	9.17 GiB	68.87 KiB
HackerNews	3.90 GiB	0.62%	2.0	7.80 GiB	4.92 KiB
YoutubeSubtitles	3.73 GiB	0.60%	2.0	7.47 GiB	22.55 KiB
PhilPapers	2.38 GiB	0.38%	2.0	4.76 GiB	73.37 KiB
NIH ExPorter	1.89 GiB	0.30%	2.0	3.79 GiB	2.11 KiB
Enron Emails [†]	0.88 GiB	0.14%	2.0	1.76 GiB	1.78 KiB
The Pile	825.18 GiB			1254.20 GiB	5.91 KiB

- **22 component datasets:** e.g., ArXiv, GitHub, OpenWebText2
- **Diverse domains:** biomedical, programming, law, subtitles, online forums
- **Top contributors** (by size/weight): Pile-CC (18%), PubMed Central, Books3, ArXiv, GitHub

Composition of the Pile by Category

■ Academic ■ Internet ■ Prose ■ Dialogue ■ Misc



Key Takeaways: The Pile paper

- Smaller, high-quality sources (e.g., academic, professional) outperform large-scale web-crawled data for training LLMs.
- Bias analysis (pejorative, gender, religion) shows patterns consistent with prior studies, suggesting similar ethical challenges persist.
- Contains a lot of information that's not well covered by GPT-3's dataset.
- Most effective components:
 - DM Mathematics, Enron Emails, PubMed Central have the highest positive impact on performance.
- Less effective or redundant components:
 - YoutubeSubtitles, Gutenberg (PG-19), and OpenWebText2 contribute the least, with some even showing negative impact.

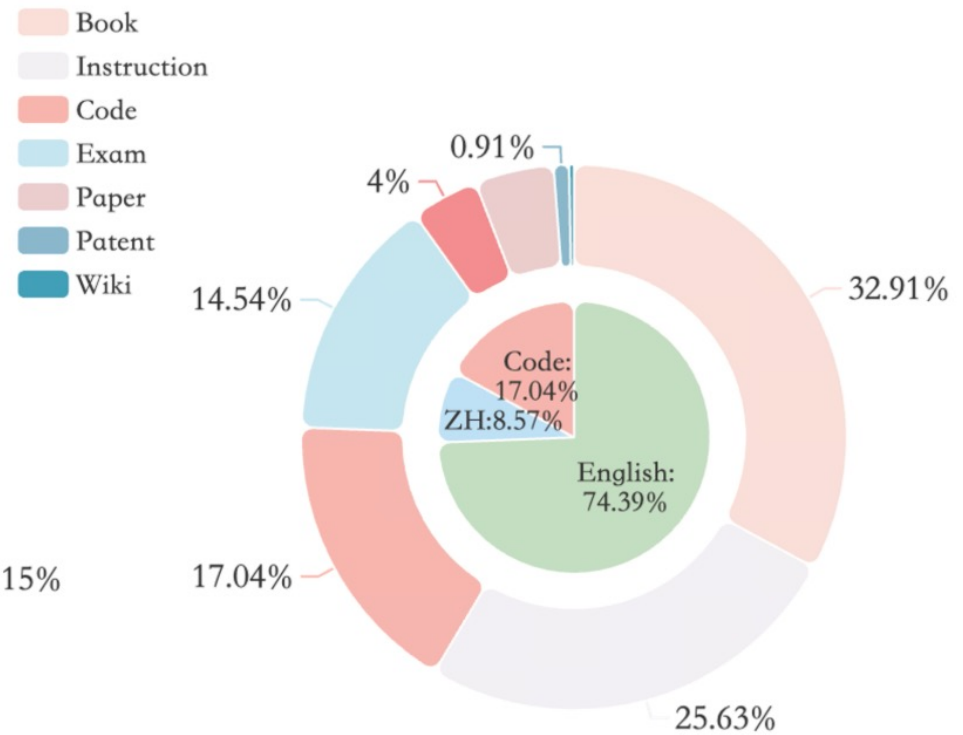
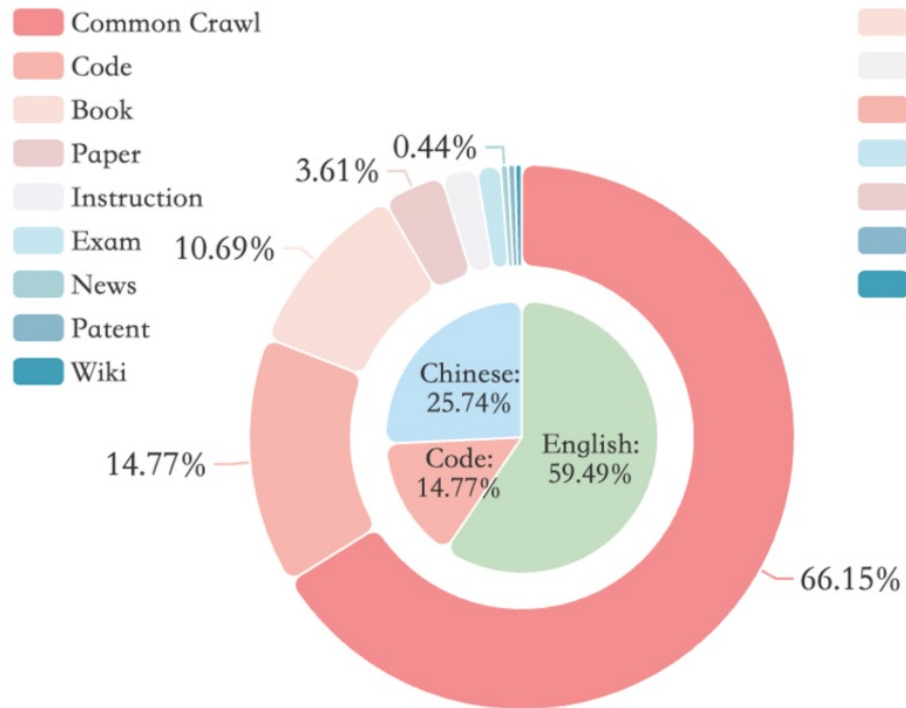
Biases in Data Representation

- Training data is **not demographically neutral** — it reflects the biases of those who produce content online.
- Internet data **overrepresents young users from developed countries**, e.g. *Your grandma isn't posting on Reddit... probably :)*
- GPT-2's data is **Reddit-based**:
 - 67% of U.S. users were men
 - 64% were aged 18–29 (Pew Research, 2016)
- Only **~9–15% of Wikipedians** are female → strong gender imbalance.

What about the elderly?

- Developed by MAP (Multimodal Art Projection), the open-source AI research group behind MAP-Neo.
- **Goal:** (1) Assemble a truly transparent, bilingual (English + Chinese) pre-training corpus. (2) Boost coverage of under-represented domains (e.g. biomedical, legal, programming, online forums).
- **Content:** 4.5 trillion tokens of cleaned text drawn from 22 heterogeneous sources, including:
 - **Web crawl:** Common Crawl (52.6 %)
 - **Programming code:** GitHub, StackExchange, etc. (22.3 %)
 - **Academic & professional:** arXiv, PubMed, legal documents, textbooks
 - **Other printed/structured:** OCR'd PDFs of books, government reports, exams

Matrix



FineWeb-Edu Dataset Overview

- Developed by **Hugging Face**, an open-source AI research collaboration.
- **Goal:** (1) Assemble a high-quality, education-focused corpus for pre-training open LLMs (2) Enhance model performance on reasoning & knowledge tasks with minimal data.
- **Content:** 1.3 trillion tokens of grade-school & middle school level web pages filtered from FineWeb heterogeneous sources
- **Balances quality & diversity by combining:**
 - Curated educational sites (K-12 lesson plans, school portals)
 - Textbook excerpts & encyclopedic articles
 - Vetted forum Q&A for informal explanations

Whose Voices Are Missing?

- Online harassment can discourage marginalized groups (e.g., queer, trans people) from contributing.
- Removes noise, but also meaningful content:
 - 22–36% of filtered LGBTQ+ mentions were benign
- Dialects like African-American English filtered 42% of the time vs. 6.2% for Standard English

Cleaning steps can systematically silence marginalized communities.

C4 Dataset – Analysis

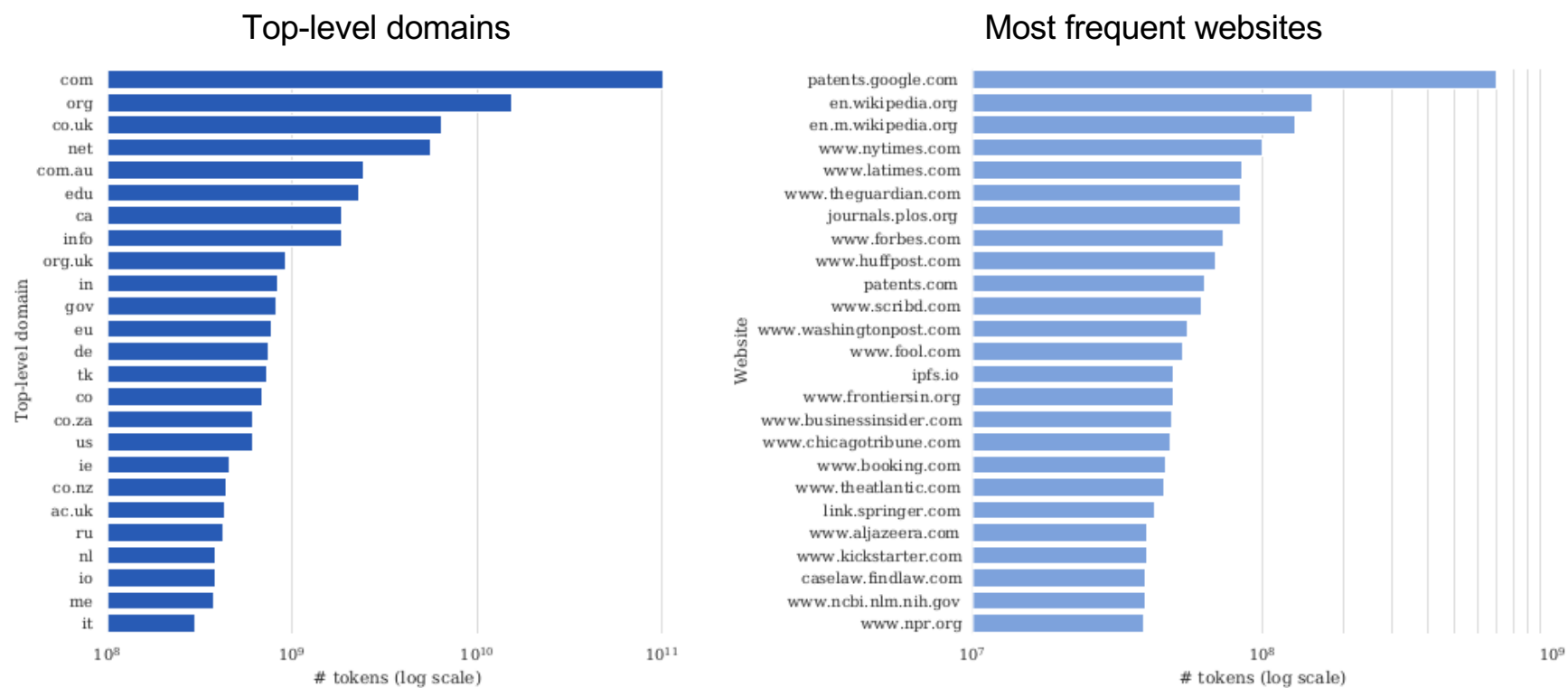
- Dodge et al. (2021) conducted an extensive analysis of the C4 dataset (used in T5 pretraining).
- Evaluated document-level quality and content characteristics
 - Captured metadata: source domain, document structure, utterance style
- Analysed included content: human vs. machine-authored, presence of social biases, contamination with evaluation data
- Noted excluded content: documents with medical/health data, or explicit demographic information, e.g. Personally identifiable information (PII)

Note: [Raffel et al. 2020](#) only provided scripts to recreate; cost thousands of dollars just to run these scripts.

C4 Dataset – Source Distribution & Quality Concerns

- A surprising amount of data from patents.google.com
- 65% pages in the Internet Archive; out of those, 92% were written in the past 10 years
- 51.3% pages are hosted in the United States; fewer from India even though lots of English speakers there
- Some patents.google.com texts are auto-generated and contain systematic errors:
 - Documents originally filed in foreign languages (e.g., Japanese) are machine-translated into English
 - Extracted via OCR, leading to transcription issues

C4 – Token Distribution by Domain



- (1) Token frequency measured in log-scale; (2) heavy dominance of US-based commercial domains and high-frequency automated sites.

What is Data Contamination?

- **Definition:** When benchmark test data appears in pretraining corpora, either partially or entirely.
- **Why it matters:** Leads to inflated benchmark performance and misleading conclusions about a model's generalization.
- **Examples**
 - XSUM (summarization): ~15.5% overlap (input + output)
 - QNLI (Wikipedia-based): up to 53.6% overlap (input only)

Types of Data Contamination

- **Input-and-output contamination**
 - Both the input and expected output appear in the training data.
 - Ranges from 1.87%–24.88% (e.g., XSUM = 15.49%)
- **Input-only contamination**
 - Only the input is present in the training set.
 - Up to 53.6% (e.g., QNLI, based on Wikipedia)
- **Which type is more harmful?**
 - *Input-and-output*: the model can memorize the entire input-output pair, severely inflating benchmark scores
 - *Input only*: still problematic but generally less severe than full input-output overlap

Real Example – XSUM Contamination

Example from the [XSum](#) summarization dataset:

- **Input:** *The 48-year-old former Arsenal goalkeeper played for the Royals for four years. He was appointed youth academy director in 2000 and has been director of football since 2003. **A West Brom statement said: “He played a key role in the Championship club twice winning promotion to the Premier League in 2006 and 2012.***
- **Output:** *West Brom have appointed Nicky Hammond as technical director, ending his 20-year association with Reading.*

Representational Harms: Sentiment Bias

- Analysis examines co-occurrence of identity terms (e.g., Jewish, Arab) with sentiment-bearing words (e.g., successful, dangerous).
- Example findings:
 - Jewish: 73.2% positive sentiment
 - Arab: 65.7% positive sentiment (7.5% difference)
- News outlet variation:
 - New York Times: 4.5% gap
 - Al Jazeera: no difference

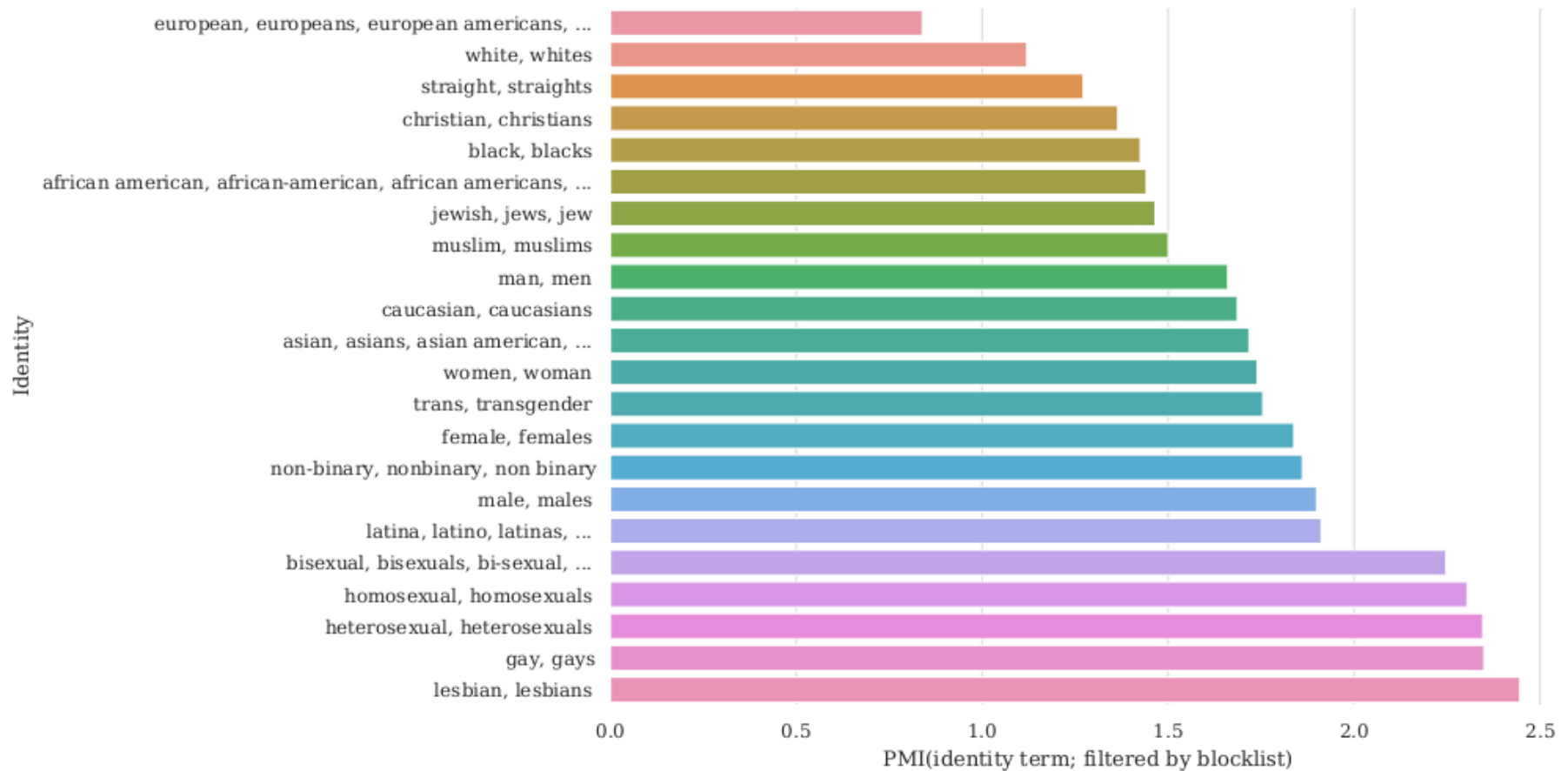
Even subtle sentiment biases can skew the representation of identities in model training data.

Representational Harms - Filtering Bias

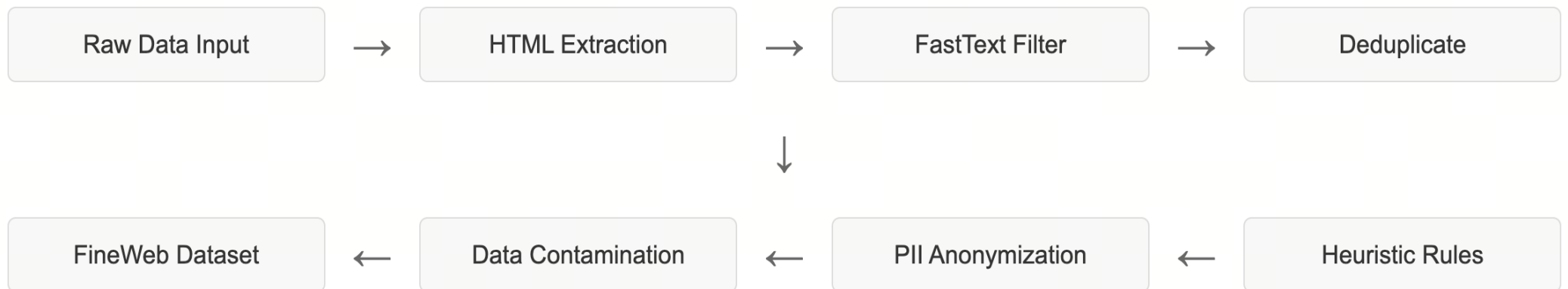
- Dialect-based bias:
 - African American English (AAE): 42% filtered
 - Hispanic-aligned English: 32%
 - White American English: only 6.2%

Filtering heuristics can reinforce systemic erasure of minority voices, even when content is benign.

Bias in Filtering: Which Groups Are Affected Most?



Common Data Filtering Pipeline



Base Filter

- **Goal:** Perform initial quality screening on raw web text to remove obviously low-quality content and non-English text, establishing a clean foundation dataset for subsequent processing.
- **Language Identification:** Use FastText classifier to filter non-English content, retaining text with English probability ≥ 0.65
- **URL Blacklist:** Remove adult content, spam websites and other inappropriate sources
- **Quality Filtering:** Apply MassiveText rules to filter overly short text, repetitive content and malformed documents
- **Basic Cleaning:** Remove obvious template text, navigation menus and advertising content

Deduplication

- **Goal:** Eliminate duplicate content in the dataset to prevent model over-memorization of repeated text during training, improving data diversity and training efficiency.
- **MinHash Algorithm:** Use 5-grams and 112 hash functions for fuzzy deduplication
- **Similarity Threshold:** Set 75% similarity threshold to identify near-duplicate documents
- **Individual Snapshot Processing:** Deduplicate each Common Crawl snapshot independently to avoid over-removal
- **Clustering Deduplication:** Use transitive clustering to ensure only one copy remains in similar document groups

Heuristic Rules

- **Goal:** Develop precise heuristic rules based on data analysis to further enhance text quality, filtering documents with formatting anomalies and poor content quality.
- **Punctuation Ratio:** Filter documents with $<12\%$ lines ending in punctuation to ensure content completeness
- **Duplicate Line Control:** Remove documents with $>10\%$ duplicate line character ratio to reduce templated content
- **Short Line Filtering:** Filter documents with $>67\%$ short lines (<30 characters) to ensure substantial content
- **Data-Driven:** Determine threshold parameters through comparative analysis of high vs low quality data

PII Anonymization

- **Goal:** Protect personal privacy by removing or anonymizing personally identifiable information, ensuring the dataset complies with privacy protection requirements and legal regulations.
- **Email Addresses:** Use regex patterns to identify and anonymize all email addresses
- **IP Addresses:** Detect and mask public IP address information
- **Pattern Matching:** Rule-based personal information identification and replacement
- **Privacy Compliance:** Ensure dataset release meets data protection regulatory requirements

Neural Scaling law

- **Background & Motivation**

- Model sizes exploded from millions → billions of parameters, driving training costs and runtimes through the roof
- Lacked quantitative guidance on when to invest in larger models versus more data or compute

- **Key Finding**

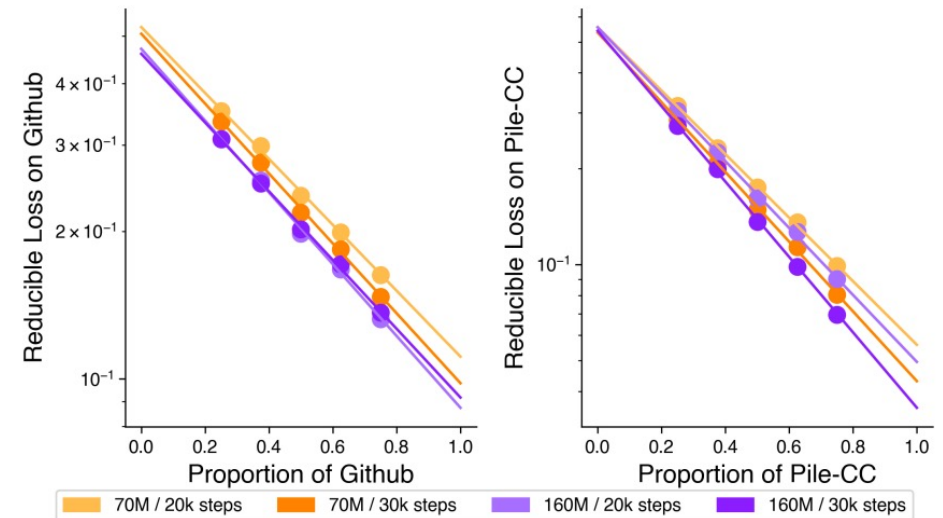
- Test loss L follows power-law relationships with parameters N , data D , and compute C

- For each modality, they fixed one of the two C , N , and varying the other one (D is varied along using $D = C/6N$), the achievable test loss satisfies

$$L = L_0 + \left(\frac{x_0}{x}\right)^\alpha$$

Data Mixing Scaling Laws

- Introduces an exponential-form law to predict domain-specific loss L as a function of data proportions t
- Select representative data domains and design a set of mixing ratios.
- Train small proxy models, record performance per domain.



Summary

- The total amount of data out there (web, private data) is massive.
- Training on “all of it” (even Common Crawl) doesn’t work well (not effective use of compute).
- Filtering / curation (OpenWebText, C4, GPT-3 dataset) is needed, but can result in biases.
- Curating non-web high-quality datasets is promising (The Pile).
- Important to carefully document and inspect these datasets.